WILFRID LAURIER UNIVERSITY

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Course Syllabus

CP400V: Cloud Computing
Computer Science & Physics, Faculty of Science, Waterloo Campus
Fall 2026|12 Week Course

I acknowledge that in Kitchener, Waterloo, Cambridge and Brantford we are on the traditional territory of the Neutral, Anishnaabe, and Haudenosaunee peoples.

Course Staff

Instructor: Dr. Shimaa Abdelmeguid
Contact Information: sabdelmeguid@wlu.ca (your email subject should include "**CP400V**")
Office Hours: Tuesday and Thursday, 4:00 PM – 5:00 PM (+15488893942 via Microsoft Teams)

Graduate Teaching Assistant:
Name: Adarsha Ghimire
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Calendar Description

This course provides students with a comprehensive understanding of cloud computing from a conceptual, technological, and architectural perspective. Students will explore cloud-enabling technologies, containerization, security mechanisms, cloud architectures, and cost/pricing models, with emphasis on practical design and application.

Prerequisite:
3 lecture hours
Credit: 0.50

Lecture Schedule:

Section	Location	Date&time
A	Bricker Academic Building: BA111	TR 2:30 PM – 3:50 PM

Midterm Exam Date (Two Midterms)

(I) October 29, 2026 (In Class)
(II) November 26, 2026 (In Class)

Final Exam Date: No Final Exam

Course Overview and Approach

This course introduces the fundamental concepts, technologies, security mechanisms, management mechanisms, and architectures of cloud computing. Students gain practical experience with cloud resources, virtualization, containers, deployment, security, monitoring, and scalability. The course concludes with the investigation and design of cloud architectures, emphasizing the selection and combination of architectural solutions to address real-world requirements and trade-offs. The course uses lectures, hands-on laboratories, case studies, and project-based learning.

Course Learning Outcomes

By the end of the course, the students should be able to:

1. Explain fundamental cloud computing concepts, models, and technologies.
2. Deploy and manage cloud-based resources and applications.
3. Apply cloud computing mechanisms for resource, SLA, security, and service management.
4. Analyze and address security requirements in cloud environments.
5. Monitor and evaluate cloud application performance and scalability.
6. Analyze fundamental, advanced, and specialized cloud architectures.
7. Design and justify cloud solutions based on functional and non-functional requirements.
8. Evaluate architectural trade-offs involving performance, scalability, reliability, security, and cost.
9. Communicate cloud computing solutions through technical documentation, diagrams, presentations, and demonstrations.

Course Tools and Learning Materials

- **Required Textbook:**

Item	**Required or Optional**	**Cost**	**Current or Most Recent Cost***	**Notes/Features (i.e. physical copy, virtual copy, second-hand availability, special features, etc.)**
Thomas Erl, *Cloud Computing: Concepts, Technology, Security & Architecture*, 2nd Edition, Pearson, 2024.	Optional	None	**$65.61**	**THE OPT OUT DEADLINE FOR Fall 2026** (12 week course) is September 24, 2026. Please email digitaltextaccess@wlu.ca if you have any questions related to your DTA course. If you have questions about your student account, or fees, please visit Service Laurier directly.

- **MyLearningSpace:**
  The MyLearningSpace website accompanying this course will contain lecture slides, hands-on labs, quizzes, and other resources related to the course.

Course Topics and Tentative Weekly Schedule

Throughout the academic year, it is important to plan ahead and pay attention to Important Dates and Deadlines on the University Calendar. Take note of important dates such as reading week, exam periods, statutory holidays as well as administrative deadlines such as course drop dates and tuition due dates.

Week #	Topic	Reference	Assessments
1 (Sep 10 & Sep 17)	Understanding Cloud Computing & Case Studies	Chapter 2 & 3	
2 (Sep 22 & Sep 24)	Fundamental Concepts and Models, Cloud-Enabling Technology	Chapter 4& 5	
3 (Sep 29 & Oct 1)	Cloud-Enabling Technology, Understanding Containerization	Chapter 5 & 6	Quiz 1- (content covered chapter 3, 4 & 5) **Due Date: Oct 02 at 11:59 PM**
4 (Oct 6 & Oct 8)	Understanding Containerization, Common Containerization Technologies	Chapter 6	
5 (Oct 20 & Oct 22)	Cloud Infrastructure Mechanisms, Specialized Cloud Mechanisms	Chapter 8 & 9	Quiz 2- (content covered chapter 6) **Due Date: Oct 23 at 11:59 PM**
6 (Oct 27 & Oct 29)	Cloud Infrastructure Mechanisms, Specialized Cloud Mechanisms **Midterm (I) Oct 29 (In Class)**	Chapter 8 & 9	
7 (Nov 3 & Nov 5)	Understanding Cloud Security and Cybersecurity	Chapter 7	Quiz 3- (content covered chapter 8 & 9) **Due Date: Nov 06 at 11:59 PM**
8 (Nov 10 & Nov 12)	Understanding Cloud Security and Cybersecurity	Chapter 7	
9 (Nov 17 & Nov 19)	Cloud Security and Cybersecurity Access-Oriented Mechanisms, Cloud Security and Cybersecurity Data-Oriented Mechanisms	Chapter 10 & Chapter 11	
10 (Nov 24 & Nov 26)	Cloud Management Mechanisms **Midterm (II) Nov 26 (In Class)**	Chapter 12	Quiz 4- (content covered chapter 7, 10 & 11)

			Due Date: Nov 25 at 11:59 PM
11 (Dec 1 & Dec 3)	Cloud Computing Architecture	Chapter 13-15	
12 (Dec 8)	Cloud Computing Architecture	Chapter 13-15	

Assessment and Grading

Assessment	**Weighting**
Quizzes	20%
Midterm Exam I	25%
Midterm Exam II	25%
Hands-on Lab and Final Project	30%
Total	**100%**

Quizzes

- There will be Four quizzes. Each quiz will be held according to the schedule below. Quiz will be available one day before the due date. All the quizzes carry equal weight.
- The quizzes will be conducted online through MLS and will contain true-false and/or multiple-choice questions.

Assessment Topics and Due Dates

Quizzes	**Topics**	**Due Date**
Quiz 1	chapter 3, 4 & 5	Oct 02 at 11:59 PM
Quiz 2	chapter 6	Oct 23 at 11:59 PM
Quiz 3	chapter 8 &9	Nov 06 at 11:59 PM
Quiz 4	chapter 7, 10, 11 &12	Nov 25 at 11:59 PM

Hands-on Labs and Final Project:

Individual hands-on labs will be released throughout the semester to provide practice with cloud computing concepts, technologies, and mechanisms. Labs may involve cloud resource deployment, virtualization, containers, security, networking, storage, monitoring, scalability, and cloud management. These activities are designed to reinforce lecture material and develop practical cloud computing skills.

Hands-on Labs	Topics / Chapters	Due Date
Lab 1	Chapters 3, 4 & 5: Cloud Computing Fundamentals, Models & Cloud-Enabling Technologies	Oct 09 at 11:59 PM
Lab 2	Chapter 6: Understanding Containerization	Oct 23 at 11:59 PM
Lab 3	Chapter 7: Cloud Security & Container Security	Nov 27 at 11:59 PM
Final Project	Cloud Architecture Investigation, Design & Prototype	Dec 10 at 11:59 PM

Midterm

Two midterms will follow the same format. They will consist of true/false and/or multiple-choice questions. They will be closed book and written in-person Through MyLS.

University and Course Policies

Late Submission Policy: Any assignment that is submitted after the deadline but before 24 hours past the deadline will receive a 20% grade deduction. No assignment will be accepted after 24 hours past the deadline. Remember that an attempt to submit in the last few minutes frequently fails. You can submit your work as many times as you want before the deadline, so please submit often and early. It is your responsibility to submit the right file in the right folder at the right time.

Unacceptable reasons for non-submission, incorrect submission, or late submission include, but are not limited to the following:

- My internet was not working, so please exempt me from the late submission penalty as it was only 5 minutes late.
- I experienced technical issues with my computer and couldn't submit on time. Now that the Dropbox is closed, I have attached my assignment here. Note that any assignments emailed to the instructor or a TA will be disregarded.
- I had another assignment due the same day and accidentally submitted that assignment to this course's Dropbox, or vice versa.
- I accidentally submitted the assignment questions instead of answers.
- I have proof that I did complete it in time. Remember, marks are not awarded for completing an assignment; they are awarded for successfully submitting the correctly completed work.

Due Date Extensions: In the event of extenuating circumstances, you may request an extension to an assignment due date without penalty. Requests must be made to your instructor at least 24 hours prior to the original due date of the assignment, and some substantiating documentation is often required. In the absence of substantiating documentation, the normal late penalty will apply.

Examples of legitimate reasons for assignment due date extension include:

- Incapacitating illness;
- Death in the family;
- Accommodations for religious observances must be submitted within two weeks of the start of term using the Student Request for Accommodation for Religious Observances form.

Most other reasons, apart from those mentioned above, are unacceptable. These include, but are not limited to:

- Vacations
- Varsity sports practices
- Work/volunteer schedules
- Student's forgetfulness or carelessness in planning their schedule
- Multiple assessments taking place on the same day.

Marking of Assignments and Midterm Exam: If you receive marks for the assignments or the midterm and you do not agree with the assessment for valid reasons, you have two weeks to bring the issue to the attention of the marker. If you are not satisfied with the marker's review, you may bring the issue to the instructor's attention. Two weeks after the day you received your mark, the assessment will not be reviewed, and your mark will not be eligible to be changed.

To save graders' time, you will be awarded 1-2 benefit-of-the-doubt mark(s) for each assignment and the midterm exam. If you request a regrade, the benefit-of-the-doubt mark(s) will be removed before the regrade, so it is not worth requesting a regrade for just a few marks. This excludes cases where a question was skipped during grading or when there was a counting error.

Missed Midterms: **Missed Midterm Exams:** Students who miss a midterm for a legitimate and **unpredictable** reason (e.g., sudden illness) must contact the course instructor within **72 hours** of the scheduled exam to arrange an alternate writing time. The request must be accompanied by Laurier's Illness Self-Declaration, or other appropriate supporting documentation. Unexplained or undocumented absences will result in a grade of **zero** for the missed midterm.

Students are advised to consult the Academic Calendars and Governance resources for expanded details of their rights and regulations.

Academic Calendars: Students are encouraged to review the Academic Calendar for information regarding all important dates, deadlines, and services available on campus.

Accessibility: Students requiring accessibility accommodation are advised to contact Accessible Learning. It is the responsibility of the student to review and understand the steps for Registration, as well as to take an active role in ensuring that information is communicated to the Accessible Learning Consultant and Instructor, as necessary. Students are responsible for meeting posted deadlines for registering with Accessible Learning and booking accommodated exams. Accessible Learning cannot guarantee accommodations for requests received after posted deadlines.

Plagiarism: Wilfrid Laurier University uses software that can check for plagiarism. If requested to do so by course instructors, students are required to submit their written work in electronic form and have it checked for plagiarism.

Academic Integrity: Laurier is committed to a culture of integrity within and beyond the classroom. This culture values honesty, integrity, reliability, fairness, caring, respect, accountability, responsibility and citizenship. Together, we have a shared responsibility to uphold this culture in our academic and nonacademic behaviour. It is expected that all students will follow assignment instructions and will adhere to Laurier's regulations with respect to Academic Integrity. As a Laurier student, you are responsible for familiarizing yourself with this policy and the accompanying penalty guidelines, some of which may appear on your transcript if there is a finding of misconduct.

Ignorance of Laurier's academic misconduct policy is not a defense

The relevant policy can be found at Laurier's Academic Integrity website along with resources to educate and support you in upholding a culture of integrity.

The following, but not limited to, are violations of Laurier's Academic Code of Conduct:

1. Accessing unauthorized resources (Google, chat rooms, Chegg, Reddit, etc.) for assessments.
2. Using technological aids beyond what is listed as permitted in an assessment.
3. Posting test, assignment, lab or exam questions to message boards, social media, or other unauthorized websites (ex. Chegg or CourseHero). While these third parties claim to be tutoring resources, for a fee they take your work and share it with other students; an act that constitutes an action of academic misconduct.
4. Copying the work of others by way of file or image sharing.
5. Posting test, assignment, lab or exam questions and answers in virtual "course groups" (ex. Facebook, Discord or WhatsApp).
6. Copying the work that is shared or posted in virtual "course groups" (ex. Facebook, Discord or WhatsApp).
7. Working collaboratively with others on assessments that are expected to be completed individually.
8. Copying the work of others by writing down answers that they tell you, but that you did not independently determine.
9. Backdating emails to gain more time to complete assessments.

Intellectual Property: The educational materials developed for this course, including, but not limited to, lecture notes and slides, handout materials, examinations and assignments, and any materials posted to MyLearningSpace, are the intellectual property of the course instructors. These materials have been developed for student use only and they are not intended for wider dissemination and/or communication outside of a given course. Posting or providing unauthorized audio, video, or textual material of course content to third-party websites violates instructors' intellectual property rights, and the Canadian Copyright Act. Recording lectures in any way is prohibited in this course unless specific permission has been granted by instructors. Failure to follow these instructions may be in contravention of the university's Student Non-Academic Code of Conduct and/or Code of Academic Conduct, and will result in appropriate penalties. Participation in this course constitutes an agreement by all parties to abide by the relevant University Policies, and to respect the intellectual property of others during and after their association with Wilfrid Laurier University.

Foot Patrol, the Wellness Centre, and the Student Food Bank (Recommended): The University approved the inclusion of information about select wellness and safety services and supports on campus in the course information provided to students. (Approved by Senate November 28, 2011.) Specific language (by campus) is provided below.

Multi-campus Resource:

- Good2Talk is a postsecondary school helpline that provides free, professional and confidential counselling support for students in Ontario. Call 1-866-925-5454 or through 2-1-1. Available 24-7.

Kitchener/Waterloo Resources:

- Waterloo Student Wellness Centre: 548-889-3239. The Centre supports the physical, emotional, and mental health needs of students. Located on the 2nd floor of the Student Services Building. Waterloo campus hours are Mondays and Wednesdays from 9:00 am to 7:30 pm, and Tuesday, Thursday, Friday from 9:00 am to 4:15 pm. Closed from 12-1 for lunch. Contact the centre: wellness@wlu.ca or @LaurierWellness. A limited number of same-day appointments are available each day for students with acute illness and unpredictable concerns. Offer online booking to registered Laurier students for non-urgent/routine appointments only.

- The Student Wellness Centre also offers hours in the Academic Advising Office, as well as other locations around campus (International Student Support Office, The Athletic Complex). Appointments around campus can be booked thought the Student Wellness Centre online booking.

- Waterloo Student Food Bank: All students are eligible to use this service to ensure they're eating healthy when overwhelmed, stressed or financially strained. Anonymously request a package online 24-7. All dietary restrictions accommodated.

- Waterloo Foot Patrol: 519.886.FOOT (3668). A volunteer operated safe-walk program, available Fall and Winter daily from 6:30 pm to 3 am. Teams of two are assigned to escort students to and from campus by foot or by van.

Brantford Resources:

- Brantford Student Food Bank: All students are eligible to use this service to ensure they're eating healthy when overwhelmed, stressed or financially strained. Anonymously request a package online 24-7. All dietary restrictions accommodated.

- Brantford Foot Patrol: 519-751-PTRL (7875). A volunteer operated safe-walk program, available Fall and Winter, Monday through Thursday from 6:30 pm to 1 am; Friday through Sunday 6:30 pm to 11 pm. Teams of two are assigned to escort students to and from campus by foot or by van.

- Brantford Wellness Centre: 519-756-8228, x5803. Students have access to support for all their physical, emotional, and mental health needs at the Wellness Centre. Location: Student Centre, 2nd floor. Hours: 8:30 am to 4:15 pm Monday through Friday. After hours crisis support available 24/7. Call 1-884-437-3247 (HERE247).